

Exercise 6.1

Show that the overhead required to terminate a message is never more than 2 bits, relative to the ideal message length given the probabilistic model $\mathcal{H}$, $h(\mathbf{x} \mid \mathcal{H}) = \log[1/P(\mathbf{x} \mid \mathcal{H})]$.

Solution:

The worst case scenario is when the string interval lies just inside a binary interval. Then, we can choose from two binary intervals. These binary symbols are no smaller than $\frac{P(\mathbf{x}|\mathcal{H})}{4}$ so the binary encoding has length no greater than

$$\log_2 1P(\mathbf{x} \mid \mathcal{H}) + \log_2 4 = \log_2 1P(\mathbf{x} \mid \mathcal{H}) + 2$$

Exercise

Solution: